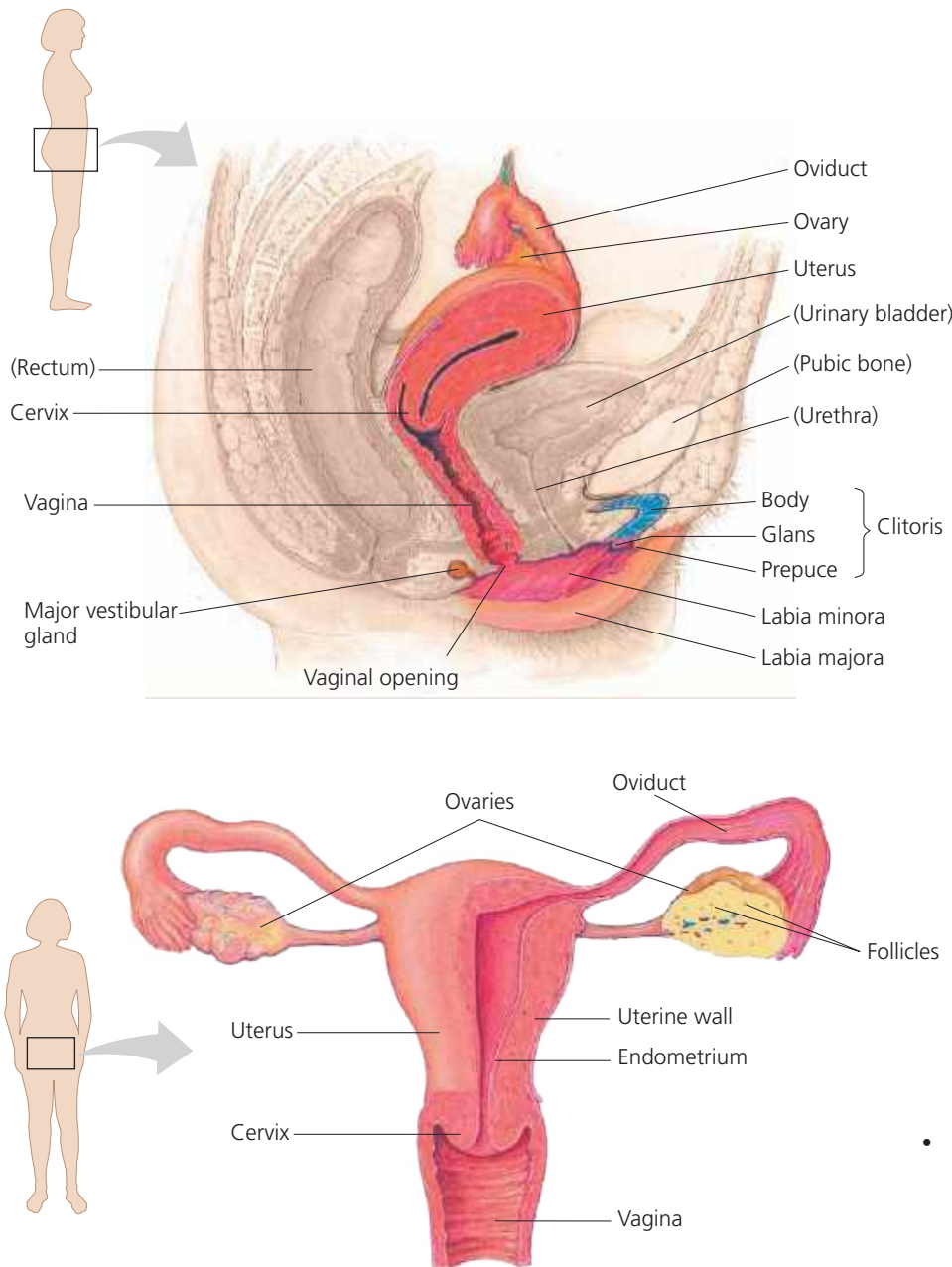


▼ **Figure 46.10 Reproductive anatomy of the human female.** Labels in parentheses identify nonreproductive structures shown for orientation purposes.



➔ **Mastering Biology Animation: Reproductive System of the Human Female**

the reproductive system, the female mammary glands are important to reproduction. Within the glands, small sacs of epithelial tissue secrete milk, which drains into a series of ducts that open at the nipple. The breasts contain connective and fatty (adipose) tissue in addition to the mammary glands.

Gametogenesis

With this overview of reproductive anatomy in mind, we turn now to **gametogenesis**, the production of gametes. In both males and females, there is a close relationship

between the gonads' structure and their function. As shown in **Figure 46.11**, there are many parallels between **spermatogenesis**—the production of sperm—and **oogenesis**—the production of oocytes (eggs). Both processes generate haploid gametes via meiotic divisions of a set of dedicated diploid cells. In addition, support cells in the gonad play an essential role in both spermatogenesis and oogenesis. However, there are also several significant differences in gametogenesis between human males and females:

- Spermatogenesis is continuous in adult males. In producing hundreds of millions of sperm each day, cell division and maturation occur throughout the seminiferous tubules. For a single sperm, the process takes about seven weeks. In contrast, oogenesis is a prolonged process in the human female. Immature eggs form in the ovary of the female embryo but do not complete their development until years, and often decades, later.
- In spermatogenesis, the four products of meiosis develop into mature gametes. In oogenesis, cytokinesis during meiosis is unequal, with almost all the cytoplasm segregated to a single daughter cell. This large cell is destined to become the egg; the other products of meiosis, smaller cells known as polar bodies, degenerate.
- Spermatogenesis occurs throughout adolescence and adulthood. In contrast, the mitotic divisions that occur in oogenesis in human females are thought to be complete before birth, and the production of mature gametes ceases at about age 50.
- Spermatogenesis produces mature sperm from precursor cells in a continuous sequence, whereas oogenesis has long interruptions.

CONCEPT CHECK 46.3

1. Why might frequent use of a hot tub make it harder for a couple to conceive a child?
2. The process of oogenesis is often described as the production of a haploid egg by meiosis, but in some animals, including humans, this description is not entirely accurate. Explain.
3. **WHAT IF?** If each vas deferens in a male were surgically sealed off, what changes would you expect in sexual response and ejaculate composition?

For suggested answers, see Appendix A.